Transportation and assignment

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Outline

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Typical mistakes

Learning objectives (1/2)

At the end of this chapter you will be able to:

- write the transportation model and its tableau, balance an instance with a dummy origin or destination, and reduce an uncapacitated transshipment instance to a transportation instance;
- recognize the bases of the transportation model as the spanning trees of a bipartite network, compute a basic solution and recognize a degenerate one;
- build an initial basic feasible solution with the north-west corner rule and the minimum cost rule;

Learning objectives (2/2)

At the end of this chapter you will be able to:

- run the potential method, with potentials, reduced costs, cycles and pivots, and certify the optimality of a solution with the dual model;
- write the assignment model, prove the theorem of König and Egerváry, and run the Hungarian algorithm, reading its potentials as an optimal dual solution.

Prerequisites

- From this module, Chapter 1 «Graphs»: trees, leaves, fundamental cycles and the rank of the node–arc incidence matrix.
- From this module, Chapter 3 «Shortest paths»: distances, and Chapter 4 «Network flow models»: the minimum cost flow model, the dummy node, integer vertices, potentials and reduced costs.
- From the module Linear Programming, Chapter 9 «Duality theory»: the dual of an LP model and the weak and strong duality theorems.

The transportation problem

Introduction: The transportation problem

- Goods stored at some origins must be shipped to some destinations, at minimum total cost
- We write an LP model for this problem and its transportation tableau, and we see that it is a minimum cost flow model on a complete bipartite network
- We see when an instance is feasible, how to balance an unbalanced instance, and how an uncapacitated transshipment instance becomes a transportation instance

Definition 1: transportation problem (1/2)

*There are m **origins** and n **destinations**. We number the origins $1, 2, \ldots, m$ and the destinations $m+1, m+2, \ldots, m+n$, and we write*

$$P = \{1, 2, \ldots, m\}, \qquad D = \{m+1, m+2, \ldots, m+n\}.$$

- *Every origin $i \in P$ has a **supply** $a_i > 0$, every destination $j \in D$ has a **demand** $d_j > 0$, and $c_{ij} \in \mathbb{R}$ is the cost of shipping one unit from i to j.*

Definition: transportation problem (2/2)

- *The instance is* ***balanced*** *if* $\sum_{i \in P} a_i = \sum_{j \in D} d_j$.

The ***transportation problem*** *asks for the amounts* $x_{ij} \geq 0$ *to ship from every origin* i *to every destination* j*, so that every origin ships its whole supply and every destination receives its demand, at minimum total cost.*

The network and the variables

- In this chapter m and n are the numbers of origins and destinations
- The underlying network has $m + n$ nodes and one arc (i, j) for every $i \in P$ and $j \in D$, that is $m\,n$ arcs: it is the **complete bipartite network** of P and D, since every arc goes from P to D
- Numbering the destinations after the origins gives every node its own number, and later its own potential
- We use one continuous variable x_{ij} for every pair $i \in P$, $j \in D$: the amount shipped from i to j

The TP model

Using these variables, an LP model for the transportation problem reads as follows:

$$\min \quad \sum_{i \in P} \sum_{j \in D} c_{ij}\, x_{ij}$$

$$\text{subject to} \quad \sum_{j \in D} x_{ij} = a_i, \quad \forall i \in P,$$

$$\sum_{i \in P} x_{ij} = d_j, \quad \forall j \in D,$$

$$x_{ij} \geq 0, \quad \forall i \in P,\ \forall j \in D.$$

This LP model is referred to as the TP model.

The TP model: objective function and constraints

Description of the objective function and constraints of the TP model:

- the objective function minimizes the total shipping cost;
- the supply constraints, one for every origin, ship the whole supply of the origin;
- the demand constraints, one for every destination, deliver the whole demand of the destination;
- finally, the last constraints define the variables of the LP model.

A minimum cost flow model

- Multiplying the demand constraints by -1 gives the flow conservation constraints of the complete bipartite network, with balances

$$b_i = a_i \text{ for } i \in P, \qquad b_j = -d_j \text{ for } j \in D,$$

  and no capacities

- The TP model is the MCF model of the row «Transportation» of the table «Special cases» of Chapter 4 «Network flow models»

Proposition 1: feasibility and integrality

An instance of the transportation problem has a feasible solution if and only if it is balanced. A balanced instance has an optimal solution, and if the supplies a_i and the demands d_j are integer, it has an integer optimal solution.

Proof (1/2)

Let $S = \sum_{i \in P} a_i$.

1. *Balance is necessary.* If $\boldsymbol{x}$ is feasible, adding up the supply constraints and the demand constraints gives the same sum of all the variables:

$$\sum_{i \in P} a_i = \sum_{i \in P} \sum_{j \in D} x_{ij} = \sum_{j \in D} d_j.$$

2. *Balance is sufficient.* If $S = \sum_{j \in D} d_j$, the amounts $x_{ij} = a_i \, d_j / S \geq 0$ are feasible: row i adds up to $a_i \sum_{j \in D} d_j / S = a_i$, and column j to $d_j \sum_{i \in P} a_i / S = d_j$.

Proof (2/2)

3. *Optimal solution.* Every feasible solution has $0 \leq x_{ij} \leq a_i$, so the feasible region is a nonempty bounded polyhedron, and a linear function attains its minimum over it at a vertex (Chapter 15 «The geometry of Linear Programming» of the module Linear Programming).

4. *Integrality.* With integer balances and no capacities, every vertex is integer by the Theorem «integer vertices» of Chapter 4.

□

The transportation tableau

- The **transportation tableau** is an $m \times n$ table with one row for every origin and one column for every destination
- The cell (i, j) holds the cost c_{ij}, or later the amount x_{ij}; the last column holds the supplies a_i and the last row the demands d_j
- A vector $\boldsymbol{x} \geq 0$ written in the tableau is feasible if and only if every row adds up to its supply and every column to its demand

Example: three warehouses and four stores (1/3)

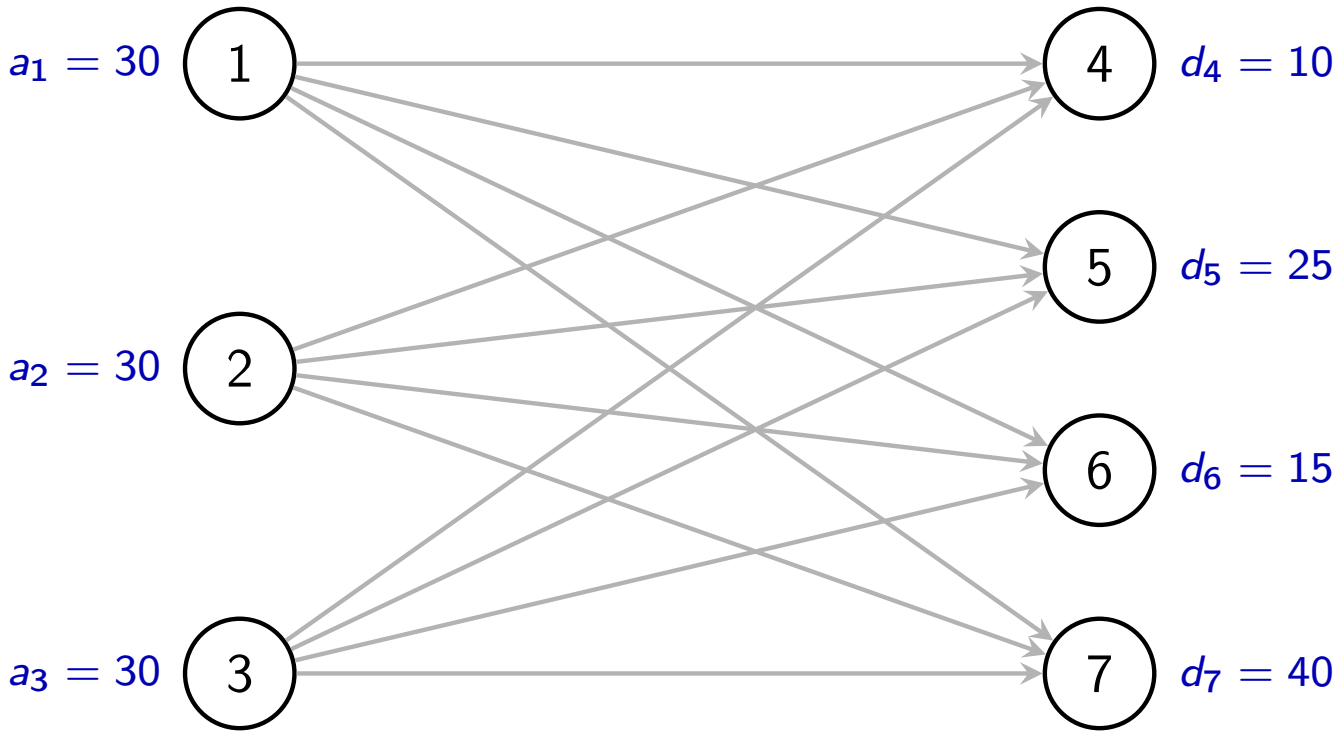
A company has three warehouses, the origins $1, 2, 3$, with 30 units each, and four stores, the destinations $4, 5, 6, 7$, with demands 10, 25, 15 and 40. The unit costs are in the tableau:

c_{ij}	4	5	6	7	a_i
1	1	9	2	3	30
2	3	4	2	2	30
3	9	1	4	6	30
d_j	10	25	15	40	90

The instance is balanced: the total supply and the total demand are both 90.

Example: three warehouses and four stores (2/3)

In the complete bipartite network every origin is joined to every destination, with $3 \cdot 4 = 12$ arcs; next to each node, its supply or its demand.



Example: three warehouses and four stores (3/3)

- The proportional amounts $x_{ij} = a_i\, d_j/90$ of the Proposition «feasibility and integrality» are feasible, for example

$$x_{14} = \frac{30 \cdot 10}{90} = \frac{10}{3},$$

  but fractional

- The vertices of the feasible region are integer

We use this instance as the example of the transportation problem in the whole chapter.

Unbalanced instances: a dummy destination

- If the total supply exceeds the total demand and every origin ships *at most* its supply, we add a **dummy destination** with demand

$$\sum_{i \in P} a_i - \sum_{j \in D} d_j$$

  and cost 0 from every origin

- The amount shipped to it is the unused supply of the origin: this is the dummy node of Chapter 4, as a column of the tableau

Unbalanced instances: a dummy origin

- If the total demand exceeds the total supply and a destination may receive less than its demand, we add a **dummy origin** with supply

$$\sum_{j \in D} d_j - \sum_{i \in P} a_i$$

- The cost of its cell $(\cdot, j)$ is the penalty for each unit of unmet demand of j, or 0 if there is none

Example: a dummy destination

In the example of the three warehouses the supply of warehouse 1 grows to $a_1 = 40$: the total supply is 100 and the total demand 90.

- We add the dummy destination 8 with demand $d_8 = 100 - 90 = 10$, and the cells $(1, 8)$, $(2, 8)$, $(3, 8)$ with cost 0: the tableau has a fifth column.
- The optimal solution, computed with Gurobi, is unique:

$$\tilde{x}_{14} = 10,\ \tilde{x}_{16} = 15,\ \tilde{x}_{17} = 10,\ \tilde{x}_{18} = 5,$$
$$\tilde{x}_{27} = 30,\ \tilde{x}_{35} = 25,\ \tilde{x}_{38} = 5,$$

  with cost 155.
- Warehouses 1 and 3 keep 5 units each, and warehouse 2 ships all its supply.

Transshipment as transportation

- In a transshipment instance the units may pass through intermediate nodes
- When the arcs have no capacities, every unit can travel along a shortest path, and only the distances between supply and demand nodes matter
- The next proposition makes this precise: an uncapacitated transshipment instance is a transportation instance with the distances as costs

Proposition 2: transshipment as transportation

Let $G = (N, A)$ be a network with balances b_i, zero lower bounds, no capacities and costs $c_{ij} \geq 0$. Let $P = \{i \in N : b_i > 0\}$ and $D = \{j \in N : b_j < 0\}$, and assume that every $j \in D$ is reachable from every $i \in P$, with distance $\tilde{d}(i, j)$ for the lengths c_{ij}. Then the minimum cost of a feasible flow equals the optimal value of the transportation instance with origins P, supplies $a_i = b_i$, destinations D, demands $d_j = -b_j$ and costs $\tilde{d}(i, j)$.

Proof (1/4)

We compare the two optimal values.

1. *From shipments to flows.* Let $\boldsymbol{y}$ be feasible for the transportation instance. Sending y_{ij} units along a shortest path from i to j, for every pair, gives a flow $\boldsymbol{x} \geq 0$:
   - node $i \in P$ sends $\sum_{j \in D} y_{ij} = b_i$ units, node $j \in D$ receives $-b_j$, the other nodes pass on what they receive.

   Its cost is $\sum_{i,j} \tilde{d}(i,j)\, y_{ij}$, so the minimum flow cost is at most the transportation optimum.

Proof (2/4)

2. *From flows to shipments.* We show by induction on the number of arcs with $x_{ij} > 0$ plus the number of nodes with $b_i \neq 0$ that every feasible flow $\boldsymbol{x}$ costs at least $\sum_{i,j} \tilde{d}(i,j)\, y_{ij}$ for some feasible $\boldsymbol{y}$. If $\boldsymbol{b} = 0$, take $\boldsymbol{y} = 0$.

3. Otherwise start at a node i_0 with $b_{i_0} > 0$: its outflow is positive, so it has a leaving arc with positive flow. A node v with $b_v \geq 0$ reached along such an arc has a positive inflow, hence a positive outflow. Follow these arcs until a node repeats or a node j_0 with $b_{j_0} < 0$ is reached.

Proof (3/4)

4. If a node repeats, the arcs form a directed cycle C with positive flows. Subtracting their minimum ε along C keeps the flow feasible, does not increase the cost, since $c \geq 0$, and empties an arc: the induction applies.

5. Otherwise we have a path Q from i_0 to j_0. Let $\varepsilon > 0$ be the minimum of b_{i_0}, $-b_{j_0}$ and the flows on Q. Subtracting ε along Q, and from b_{i_0} and $-b_{j_0}$, gives a feasible flow $\boldsymbol{x}'$ for the new balances, with fewer positive arcs or nonzero balances, and

$$\boldsymbol{c}'\boldsymbol{x} = \boldsymbol{c}'\boldsymbol{x}' + \varepsilon\, c(Q) \geq \boldsymbol{c}'\boldsymbol{x}' + \varepsilon\, \tilde{d}(i_0, j_0).$$

Proof (4/4)

- By induction
$$\boldsymbol{c}'\boldsymbol{x}' \geq \sum \tilde{d}(i,j)\, y'_{ij}$$
- Adding ε to $y'_{i_0 j_0}$ gives the required $\boldsymbol{y}$:
$$\boldsymbol{c}'\boldsymbol{x} \geq \sum \tilde{d}(i,j)\, y'_{ij} + \varepsilon\, \tilde{d}(i_0,j_0) = \sum \tilde{d}(i,j)\, y_{ij}$$

□

Example: transshipment without capacities (1/3)

We take the reference network of Chapter 4, with the same balances $\boldsymbol{b} = (5, 2, 0, -3, 0, -4)$ and costs, and remove the capacities.

- The origins are the supply nodes $1, 2$ and the destinations the demand nodes $4, 6$; we keep the names of the network.

- The distances are

$$\tilde{d}(1,4) = 7 \text{ along } 1,2,4, \quad \tilde{d}(1,6) = 8 \text{ along } 1,3,5,6,$$

$$\tilde{d}(2,4) = 5 \text{ along } 2,4, \quad \tilde{d}(2,6) = 10 \text{ along } 2,4,6.$$

Example: transshipment without capacities (2/3)

- The 2×2 transportation instance has $a_1 = 5$, $a_2 = 2$, $d_4 = 3$, $d_6 = 4$
- With $y_{24} = t \in [0, 2]$ the other amounts are $y_{14} = 3 - t$, $y_{16} = 2 + t$, $y_{26} = 2 - t$, and the cost is

$$7(3 - t) + 8(2 + t) + 5t + 10(2 - t) = 57 - 4t$$

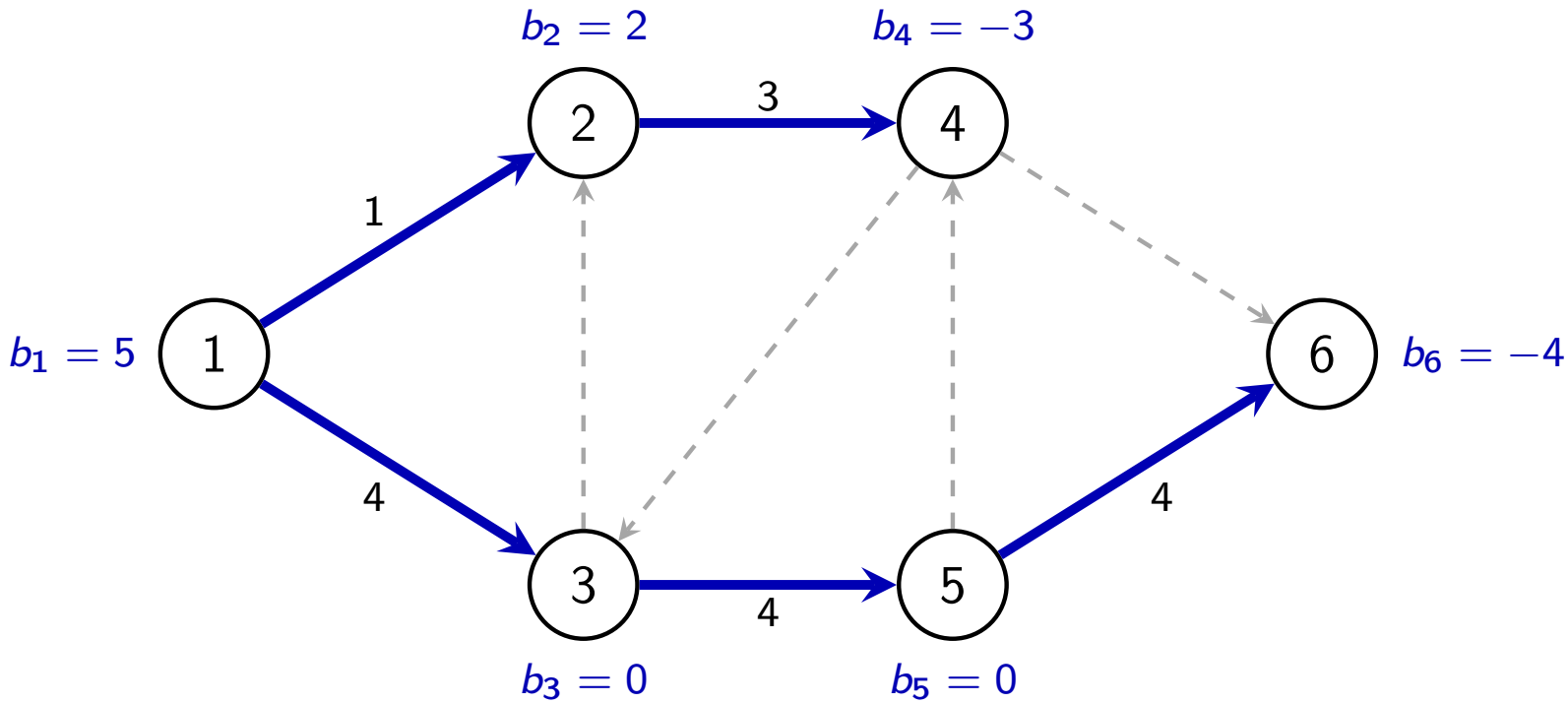
- The minimum is for $t = 2$:

$$y_{14} = 1, \quad y_{16} = 4, \quad y_{24} = 2, \quad y_{26} = 0,$$

with cost 49

Example: transshipment without capacities (3/3)

Along the shortest paths this gives the flow $x_{12} = 1$, $x_{13} = 4$, $x_{24} = 3$, $x_{35} = 4$, $x_{56} = 4$ (blue), of cost 49, the optimum computed by Gurobi.



With the capacities of Chapter 4 this flow is not feasible, since the arc $(5, 6)$ has capacity 2, and the optimum is 60.

Summary

The transportation problem:

- it ships the supplies of m origins to the demands of n destinations at minimum cost
- it is a minimum cost flow model on the complete bipartite network

Feasibility:

- an instance is feasible exactly when it is balanced
- a dummy origin or destination balances it

Transshipment:

- without capacities, it is a transportation instance with the distances as costs

Basic solutions

Introduction: Basic solutions

- The optimal solutions of an LP model are found among its vertices, that is among its basic feasible solutions
- In this section we see that the bases of the TP model have a simple form: they are the spanning trees of the complete bipartite network
- This is the key to the potential method: a basic solution is computed leaf by leaf, directly in the tableau

The constraint matrix

- The column of x_{ij} in the constraints of the TP model has two entries equal to 1, in the row of the supply of i and in the row of the demand of j
- Changing the sign of the demand rows gives the incidence matrix $\boldsymbol{B}$ of the complete bipartite network
- This changes neither the rank nor the linear dependences of the columns

Proposition 3: rank of the transportation model

The $m+n$ constraints of the TP *model have rank $m+n-1$. In a balanced instance each constraint is implied by the others.*

Proof

- Every origin is joined to every destination, so the complete bipartite network is connected, and its incidence matrix has rank $m + n - 1$ by the Theorem «rank of the incidence matrix» of Chapter 1.

- The sum of the supply rows minus the sum of the demand rows is zero, since both sums contain every variable once, and the same combination of the right-hand sides is

$$\sum_{i \in P} a_i - \sum_{j \in D} d_j = 0 :$$

any constraint is a combination of the others.

□

Definition 2: basis and basic solution

*A **basis** of the* TP *model is a set T of $m+n-1$ cells whose arcs form a spanning tree of the complete bipartite network; its cells are the **basic cells**.*

*A **basic solution** of T is a solution of the supply and demand constraints with $x_{ij} = 0$ for every cell outside T.*

*It is a **basic feasible solution** if $\mathbf{x} \geq 0$, and it is **degenerate** if some basic cell has $x_{ij} = 0$.*

Theorem 1: bases are spanning trees

(a) *A set of $m + n - 1$ cells has linearly independent columns if and only if its arcs form a spanning tree.*

(b) *In a balanced instance, every basis T has exactly one basic solution. It is found by fixing, again and again, the cell of a leaf of the tree, and it is integer when the supplies and the demands are integer.*

Proof (1/3)

- Part (a) is the Corollary «bases of the incidence matrix» of Chapter 1, since the sign of the demand rows does not change the dependences of the columns.

For part (b):

1. *Uniqueness.* Two basic solutions of T differ by a vector $\boldsymbol{y}$ that is zero outside T and satisfies the constraints with zero right-hand sides; the columns of T are independent by (a), so $\boldsymbol{y} = 0$.

Proof (2/3)

2. *Construction.* The tree has a leaf v (Lemma «leaves» of Chapter 1), with only one cell e of T. The constraint of v contains no other basic cell, so x_e equals the supply or demand of v.

3. We remove v and e, and subtract x_e from the supply or demand of the other endpoint of e. The remaining cells form a spanning tree of the remaining nodes, and the remaining supplies and demands still have equal sums.

Proof (3/3)

4. When a single cell is left, both its endpoints are leaves, and the equal sums give the same value for both constraints.

   Every value is obtained by sums and differences of the data, so it is integer for integer data.

□

Proposition 4: vertices are basic

Every vertex $\bar{\boldsymbol{x}}$ *of the feasible region of the* TP *model is a basic feasible solution: its cells with* $\bar{x}_{ij} > 0$ *form a forest, contained in a basis* T*, and* $\bar{\boldsymbol{x}}$ *is the basic solution of* T*.*

Proof (1/2)

Let F be the cells with $\bar{x}_{ij} > 0$.

1. The columns of F are linearly independent, by the first step of the proof of the Theorem «integer vertices» of Chapter 4: here the cells strictly between their bounds are those of F.

   A cycle has dependent columns (Lemma «columns of a cycle» of Chapter 1), so F is a forest.

Proof (2/2)

2. The network is connected: while the forest has two or more components, some arc joins one of them to another one, and adding it creates no cycle (Lemma «adding an edge» of Chapter 1). We end with a spanning tree $T \supseteq F$.

3. $\bar{\boldsymbol{x}}$ is zero outside $F \subseteq T$, so it is the basic solution of T, unique by the Theorem «bases are spanning trees».

□

Bases in the tableau

- A cycle of the bipartite network alternates origins and destinations
- In the tableau it is a closed path through cells of the set, moving alternately along a row and along a column
- A set of cells is a basis if it has $m + n - 1$ cells and no such closed path
- A basic feasible solution has at most $m + n - 1$ positive amounts; it has fewer when it is degenerate

Example: bases of the transportation instance (1/2)

In the example of the three warehouses a basis has $m + n - 1 = 3 + 4 - 1 = 6$ cells.

- The cells $(1, 4)$, $(1, 5)$, $(2, 4)$, $(2, 5)$, $(3, 6)$, $(3, 7)$ are not a basis
- The cells $(1, 4)$, $(2, 4)$, $(2, 5)$, $(1, 5)$ form the cycle $1, 4, 2, 5, 1$, and indeed

$$\boldsymbol{A}_{14} - \boldsymbol{A}_{24} + \boldsymbol{A}_{25} - \boldsymbol{A}_{15} = 0$$

  for their columns $\boldsymbol{A}_{ij}$

Example: bases of the transportation instance (2/2)

- The cells $(1,4)$, $(1,5)$, $(1,6)$, $(1,7)$, $(2,7)$, $(3,7)$ form a spanning tree
- Its leaves $2,3,4,5,6$ fix

$$x_{27} = 30, \quad x_{37} = 30, \quad x_{14} = 10, \quad x_{15} = 25, \quad x_{16} = 15$$

- Then the row of origin 1 gives

$$x_{17} = 30 - 10 - 25 - 15 = -20.$$

- The basic solution exists, but it is not feasible

Summary

Rank:

- the constraints of the TP model have rank $m + n - 1$

Bases:

- a basis is a spanning tree of the bipartite network, with $m + n - 1$ cells
- its basic solution is found leaf by leaf, and it is integer for integer data

Vertices:

- every vertex is a basic feasible solution

Initial basic feasible solutions

Introduction: Initial basic feasible solutions

- The potential method starts from a basic feasible solution: we need a way to build one
- We fill the tableau one cell at a time, crossing out one row or one column at every step
- Two rules choose the next cell: the north-west corner rule and the minimum cost rule

Active lines and residuals

- A basic feasible solution is built by filling the tableau one cell at a time
- A row or a column is **active** until it is crossed out
- The residual supply $\bar{a}_i$ and the residual demand $\bar{d}_j$ are what is left to ship and to receive
- The two rules below differ only in the choice of the next cell

Algorithm: Initial basic feasible solution

Input : a balanced instance with supplies a_i, demands d_j and costs c_{ij}

Output : a basis T and its basic feasible solution $\bar{\boldsymbol{x}}$

1 **Step 0 (initialization).** All rows and columns are active; $\bar{a}_i := a_i$, $\bar{d}_j := d_j$, $T := \emptyset$.

2 **Step 1 (choice of the cell).** Choose a cell (i, j) with active row and column:

3 *north-west corner rule*: the cell with the smallest i and the smallest j;

4 *minimum cost rule*: the cell with the smallest c_{ij} (the smallest i, then j, on ties).

5 **Step 2 (amount).** Set $\bar{x}_{ij} := \min\{\bar{a}_i, \bar{d}_j\}$, add (i, j) to T, and subtract $\bar{x}_{ij}$ from $\bar{a}_i$ and from $\bar{d}_j$.

6 **Step 3 (crossing out).** If $\bar{a}_i = 0$ and another row is active, cross out row i; otherwise cross out column j.

7 If no column is active, `STOP`; otherwise go to Step 1.

Ties: a basic cell with value 0

- When $\bar{a}_i = \bar{d}_j = 0$ at the same time, only the row is crossed out
- The column stays active with residual demand 0, and a later cell of that column gets the amount 0
- This basic cell with value 0 keeps T a spanning tree, and the solution is degenerate

Theorem 2: the initial solution is basic

With either rule, the algorithm «Initial basic feasible solution» stops after $m+n-1$ steps, and T is a basis with basic feasible solution $\bar{\boldsymbol{x}}$.

Proof (1/3)

1. *Invariant.* The residual supplies of the active rows and the residual demands of the active columns have equal sums: true at the start, and Step 2 subtracts the same amount from both sides.

2. *A crossed line has residual* 0. A crossed row has $\bar{a}_i = 0$. A column is crossed:
   - when $\bar{a}_i > 0$, and then $\bar{x}_{ij} = \bar{d}_j$ and $\bar{d}_j = 0$,
   - or when row i is the only active row with $\bar{a}_i = 0$, and then every active column has residual 0 by the invariant.

Proof (2/3)

3. *Counting.* A row is crossed out only if another row is active, so one row r is never crossed out, and the algorithm stops when the n columns are crossed out.

   At the stop the active rows have residual 0 by the invariant, and a row that reaches 0 without being crossed out is the only active row.

   Hence the steps are $(m-1)+n = m+n-1$, one cell each, all different, since a cell is chosen in two active lines and one of them is then crossed out.

Proof (3/3)

4. *Feasibility.* At the stop every row and every column has residual 0, so $\bar{\boldsymbol{x}} \geq 0$ satisfies all the constraints.

5. *Tree.* Let L_t be the line crossed at step t: no later cell lies in L_t.

   Backwards from the last step, the cells of steps $t, t+1, \ldots$ form a tree: the cell of step t joins the new node L_t to its other line, which is still active, hence a node of the later cells or the row r.

   At $t = 1$ the tree spans all the $m + n$ nodes.

□

Example: the north-west corner rule (1/3)

On the example of the three warehouses the north-west corner rule starts at the cell $(1, 4)$ and moves right or down. Its steps are:

step	cell	amount	crossed out
1	$(1, 4)$	10	column 4
2	$(1, 5)$	20	row 1
3	$(2, 5)$	5	column 5
4	$(2, 6)$	15	column 6
5	$(2, 7)$	10	row 2
6	$(3, 7)$	30	row 3, column 7

Example: the north-west corner rule (2/3)

The basic feasible solution, with the basic cells in blue, is

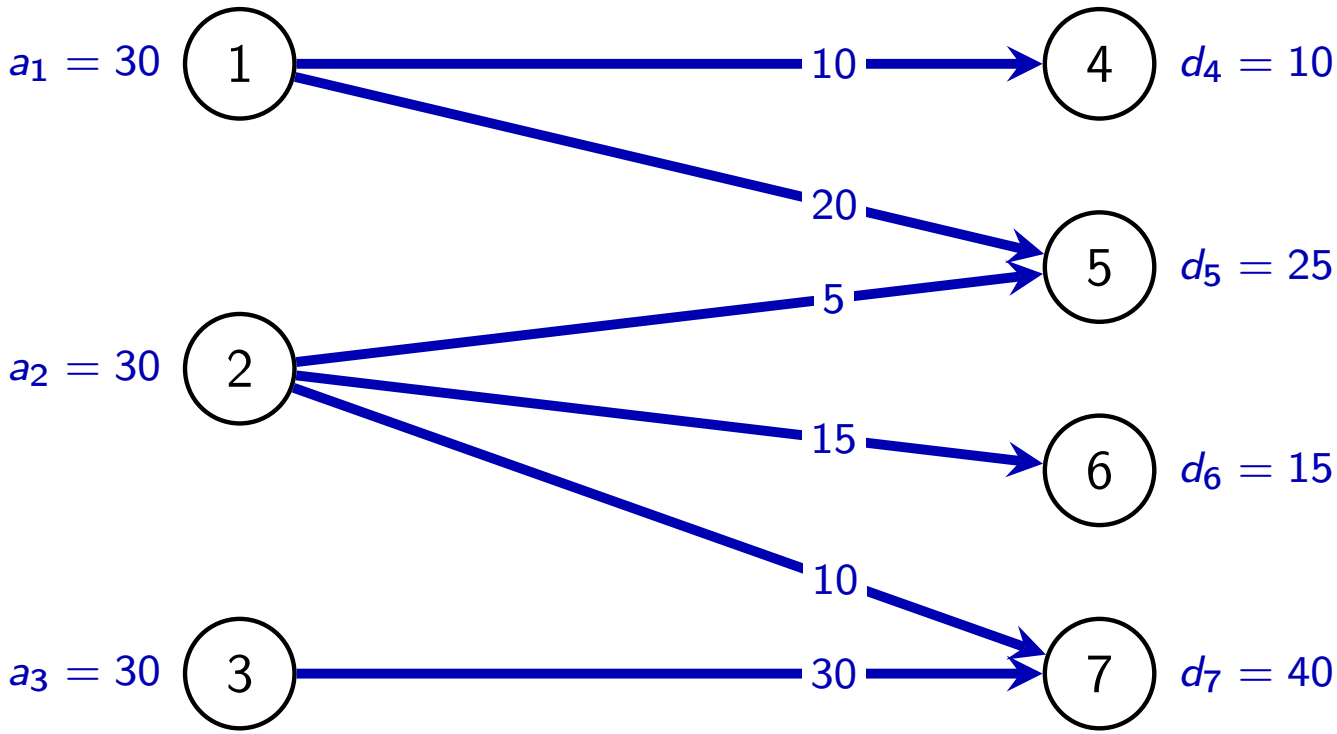
$\bar{x}_{ij}$	4	5	6	7	a_i
1	10	20			30
2		5	15	10	30
3				30	30
d_j	10	25	15	40	

Its cost is

$$1 \cdot 10 + 9 \cdot 20 + 4 \cdot 5 + 2 \cdot 15 + 2 \cdot 10 + 6 \cdot 30 = 440.$$

Example: the north-west corner rule (3/3)

The six basic cells form the spanning tree below, with the amounts on the arcs.



The north-west corner rule ignores the costs

- The north-west corner rule looks only at the positions of the cells, never at their costs
- On this instance it ships 20 units on the cell $(1, 5)$, of cost 9, and 30 units on $(3, 7)$, of cost 6
- The minimum cost rule looks at the costs instead

Example: the minimum cost rule (1/3)

On the example of the three warehouses the minimum cost rule chooses, at every step, the cheapest cell of the active rows and columns:

step	cell	amount	crossed out
1	$(1,4)$	10	column 4
2	$(3,5)$	25	column 5
3	$(1,6)$	15	column 6
4	$(2,7)$	30	row 2
5	$(1,7)$	5	row 1
6	$(3,7)$	5	row 3, column 7

Example: the minimum cost rule (2/3)

- At step 1 the cells $(1, 4)$ and $(3, 5)$ both cost 1, and $(1, 4)$ goes first, since its row is smaller
- At step 3 the cells $(1, 6)$, $(2, 6)$, $(2, 7)$ cost 2, and $(1, 6)$ goes first
- At step 4 only column 7 is active: it takes the whole supply of row 2, then 5 units from row 1 and 5 from row 3

Example: the minimum cost rule (3/3)

The basic feasible solution is

$\bar{x}_{ij}$	4	5	6	7	a_i
1	10		15	5	30
2				30	30
3		25		5	30
d_j	10	25	15	40	

with cost

$$1 \cdot 10 + 2 \cdot 15 + 3 \cdot 5 + 2 \cdot 30 + 1 \cdot 25 + 6 \cdot 5 = 170.$$

Summary

Building a basic feasible solution:

- one cell per step and one line crossed out per step
- the result has $m + n - 1$ basic cells, and it is a basis

The two rules:

- the north-west corner rule ignores the costs
- the minimum cost rule usually starts much closer to the optimum: 170 instead of 440 on the example

The potential method

Introduction: The potential method

- From a basic feasible solution we want to reach an optimal one
- The potentials of the basis give the reduced costs of the cells: when they are all non-negative the solution is optimal, and the potentials are an optimal dual solution
- Otherwise a cell with negative reduced cost enters the basis, and the amounts move around its cycle: this is the simplex algorithm, written in the tableau

Potentials as dual variables

- We write the demand constraints as

$$-\sum_{i \in P} x_{ij} = -d_j,$$

  the conservation constraints of the destinations

- We use a free dual variable π_v, the potential of node v, for the constraint of every node $v \in P \cup D$, as in Chapter 4

The $D(\mathrm{TP})$ model

Using a free dual variable π_v for the constraint of each node $v \in P \cup D$, the dual model of the TP model is:

$$\begin{aligned} \max \quad & \sum_{i \in P} a_i \, \pi_i - \sum_{j \in D} d_j \, \pi_j \\ \text{subject to} \quad & \pi_i - \pi_j \leq c_{ij}, \quad \forall i \in P, \; \forall j \in D, \\ & \pi_v \in \mathbb{R}, \quad \forall v \in P \cup D. \end{aligned}$$

This LP model is referred to as the $D(\mathrm{TP})$ model.

Reduced costs and normalization

- With the reduced cost of Chapter 4

$$\bar{c}_{ij} = c_{ij} - \pi_i + \pi_j,$$

the dual constraint $\pi_i - \pi_j \leq c_{ij}$ reads $\bar{c}_{ij} \geq 0$

- Adding the same constant to all the potentials changes neither the reduced costs nor, in a balanced instance, the dual objective

- We normalize with $\pi_{m+n} = 0$, the potential of the last destination

Cost and reduced cost

For every $\boldsymbol{x}$ that satisfies the supply and demand constraints, the potentials add up row by row and column by column:

$$\sum_{i \in P} \sum_{j \in D} (\pi_i - \pi_j)\, x_{ij} = \sum_{i \in P} a_i\, \pi_i - \sum_{j \in D} d_j\, \pi_j,$$

and since $c_{ij} = \bar{c}_{ij} + \pi_i - \pi_j$,

$$\sum_{i \in P} \sum_{j \in D} c_{ij}\, x_{ij} = \sum_{i \in P} \sum_{j \in D} \bar{c}_{ij}\, x_{ij} + \sum_{i \in P} a_i\, \pi_i - \sum_{j \in D} d_j\, \pi_j.$$

This is the Lemma «cost and reduced cost» of Chapter 4 for the bipartite network.

Definition 3: potentials of a basis

*The **potentials of a basis** T are the values π_v, $v \in P \cup D$, such that*

$$\bar{c}_{ij} = c_{ij} - \pi_i + \pi_j = 0, \quad \forall (i,j) \in T,$$

$$\pi_{m+n} = 0.$$

- The potentials of a basis exist and are unique

Computing the potentials of a basis

- Start from $\pi_{m+n} = 0$
- A basic cell (i, j) with one known endpoint gives the other one:

$$\pi_i = \pi_j + c_{ij} \qquad \text{or} \qquad \pi_j = \pi_i - c_{ij}$$

- Each node is reached along its unique path in the tree from $m + n$, which forces its value

Theorem 3: optimality criterion

Let $\bar{\mathbf{x}}$ be the basic feasible solution of a basis T, with potentials $\boldsymbol{\pi}$. If $\bar{c}_{ij} \geq 0$ for every cell, then $\bar{\mathbf{x}}$ is optimal for the TP *model, $\boldsymbol{\pi}$ is optimal for the $D(\mathrm{TP})$ model, and*

$$\sum_{i \in P} \sum_{j \in D} c_{ij}\, \bar{x}_{ij} = \sum_{i \in P} a_i\, \pi_i - \sum_{j \in D} d_j\, \pi_j .$$

If moreover $\bar{c}_{ij} > 0$ for every cell outside T, then $\bar{\mathbf{x}}$ is the unique optimal solution.

Proof (1/2)

We use the identity «cost and reduced cost». Let

$$z_\pi = \sum_{i \in P} a_i \pi_i - \sum_{j \in D} d_j \pi_j.$$

1. For $\bar{\boldsymbol{x}}$: $\bar{c}_{ij} = 0$ on T and $\bar{x}_{ij} = 0$ outside T, so $\sum \bar{c}_{ij} \bar{x}_{ij} = 0$ and the cost of $\bar{\boldsymbol{x}}$ is z_π.

2. For every feasible $\boldsymbol{x}$: $\bar{c} \geq 0$ and $\boldsymbol{x} \geq 0$, so its cost is at least z_π; hence $\bar{\boldsymbol{x}}$ is optimal.

Proof (2/2)

The potentials satisfy the dual constraints $\pi_i - \pi_j \leq c_{ij}$, and their value z_π equals a primal cost: by weak duality they are optimal.

3. If $\bar{c}_{ij} > 0$ outside T, an optimal $\boldsymbol{x}$ has $\sum \bar{c}_{ij} x_{ij} = 0$, hence $x_{ij} = 0$ outside T: it is the basic solution of T, which is $\bar{\boldsymbol{x}}$ by the Theorem «bases are spanning trees».

□

The cycle of a cell and the pivot

- When some cell has $\bar{c}_e < 0$, the criterion fails, and we try to improve the solution
- The cell e closes a cycle with the tree of the basis: moving amounts around this cycle keeps all the rows and columns balanced
- The next lemma gives the signs of the cells of the cycle, and the next theorem the pivot

Lemma 1: the cycle of a non-basic cell (1/2)

Let T be a basis and $e = (k, l) \notin T$. The cells $T \cup \{e\}$ contain exactly one cycle C, made of e and of the path from l to k in the tree.

Lemma: the cycle of a non-basic cell (2/2)

Write C starting from e and moving first along row k, and give the sign $+$ to the cells in the odd positions (e included) and $-$ to the others.

Then every row and every column that meets C contains exactly one cell of C with sign $+$ and one with sign $-$.

Proof

- The cycle is the fundamental cycle of the arc (k, l) (Proposition «fundamental cycle» of Chapter 1)
- It visits the nodes $k, j_1, i_1, j_2, \ldots, l$ alternately in P and in D, so it has an even number of cells, and two consecutive cells share a node: the first two share the row k, the next two a column, and so on
- Every node of C lies in exactly two cells of C, which are consecutive along C, the last and the first one included: their positions have different parity, hence different signs

□

Theorem 4: the pivot (1/2)

Let $\bar{\mathbf{x}}$ be the basic feasible solution of a basis T with potentials $\boldsymbol{\pi}$, let $e \notin T$ with $\bar{c}_e < 0$, let C^+ and C^- be the cells of its cycle with sign $+$ and $-$, and let

$$\theta = \min\{\bar{x}_f : f \in C^-\},$$

attained at a cell $f^* \in C^-$. For $0 \leq t \leq \theta$, let $\mathbf{x}(t)$ be $\bar{\mathbf{x}}$ with t added on the cells of C^+ and subtracted on the cells of C^-.

Theorem: the pivot (2/2)

(a) $\boldsymbol{x}(t)$ *is feasible, and its cost is the cost of* $\bar{\boldsymbol{x}}$ *plus* $t\,\bar{c}_e$.

(b) $T' = T \cup \{e\} \setminus \{f^*\}$ *is a basis, and* $\boldsymbol{x}(\theta)$ *is its basic feasible solution.*

Proof (1/2)

1. *Feasibility.* By the Lemma «the cycle of a non-basic cell» every row and column of C gains t on one cell and loses t on another, so all the sums are unchanged. The cells of C^- keep $\bar{x}_f - t \geq 0$ since $t \leq \theta$.

2. *Cost.* The change of cost is $t\big(\sum_{C^+} c_{ij} - \sum_{C^-} c_{ij}\big)$. Every node of C lies in one cell of C^+ and one of C^-, so

$$\sum_{C^+}(\pi_i - \pi_j) = \sum_{C^-}(\pi_i - \pi_j),$$

and the change is $t\big(\sum_{C^+} \bar{c}_{ij} - \sum_{C^-} \bar{c}_{ij}\big) = t\,\bar{c}_e$: the other cells of C are basic, with $\bar{c}_{ij} = 0$.

Proof (2/2)

3. *New basis.* T' has $m + n - 1$ cells, and it has no cycle: the only cycle of $T \cup \{e\}$ is C, which contains f^*.

   An acyclic set of $m + n - 1$ arcs on $m + n$ nodes is a spanning tree (Theorem «characterizations of a tree» of Chapter 1).

   $\boldsymbol{x}(\theta)$ is zero outside T', since $x_{f^*}(\theta) = 0$, so it is the basic solution of T'.

□

Algorithm: The potential method

Input : a balanced instance with supplies a_i, demands d_j and costs c_{ij}

Output : an optimal basic feasible solution $\tilde{\boldsymbol{x}}$ and optimal potentials $\tilde{\boldsymbol{\pi}}$

1. **Step 0 (initial basis).** Build a basis T and its basic feasible solution $\bar{\boldsymbol{x}}$ with the north-west corner rule or the minimum cost rule.
2. **Step 1 (potentials).** Compute the potentials of T and the reduced costs $\bar{c}_{ij} = c_{ij} - \pi_i + \pi_j$ of the cells outside T.
3. **Step 2 (optimality test).** If $\bar{c}_{ij} \geq 0$ for every cell, `STOP`: $\bar{\boldsymbol{x}}$ is optimal.
4. **Step 3 (entering cell).** Choose a cell e with $\bar{c}_e < 0$.
5. **Step 4 (cycle).** Find the cycle of e with its signs, $\theta := \min\{\bar{x}_f : f \in C^-\}$, and a leaving cell $f^* \in C^-$ with $\bar{x}_{f^*} = \theta$.
6. **Step 5 (pivot).** Add θ on C^+, subtract θ on C^-, set $T := T \cup \{e\} \setminus \{f^*\}$, and go to Step 1.

Rules of the examples and names

- **Rules of the examples.** In Step 3 the entering cell has the most negative reduced cost (the smallest i, then j, on ties). In Step 4 the cycle is written from the entering cell moving first along its row, and the leaving cell is the first cell of C^- with $\bar{x}_f = \theta$ in row–column order.

- The method is also known as the **u-v method** or MODI (modified distribution) method, and the cycle as the **stepping stone** path: the amounts move along it like on stones across a river.

Theorem 5: termination

If $\theta > 0$ at every iteration, the potential method stops after a finite number of iterations with an optimal basic feasible solution and optimal potentials.

Proof

- By the Theorem «the pivot» every iteration with $\theta > 0$ moves to a new basis and decreases the cost by $\theta \left|\bar{c}_e\right| > 0$
- A basis determines its basic solution (Theorem «bases are spanning trees»), so a basis visited twice would have two different costs: no basis repeats
- There are finitely many sets of $m + n - 1$ cells, so the method stops, at Step 2, and the Theorem «optimality criterion» applies

□

Degenerate pivots and the simplex algorithm

- When $\theta = 0$ the pivot is **degenerate**: the basis changes, the solution and its cost do not
- The potential method is the simplex algorithm on the TP model: the basic cells are the basic variables, $\bar{c}_{ij}$ are their reduced costs, and the cycle describes how the basic variables change when x_e grows
- Hence the anti-cycling rule of Bland of Chapter 7 «Optimality conditions and pivot rules» of the module Linear Programming applies: with the first cell with $\bar{c}_{ij} < 0$ in row–column order as the entering cell, and the leaving rule above, no basis repeats even with degenerate pivots

Observation 1: The u-v convention

Many textbooks write the u-v method with $u_i + v_j = c_{ij}$ on the basic cells and $\bar{c}_{ij} = c_{ij} - u_i - v_j$.

These are the potentials of this chapter with $u_i = \pi_i$ for the origins and $v_j = -\pi_j$ for the destinations: the reduced costs are the same, and the potentials of the destinations change sign.

The potential method on the transportation instance

- We start from the north-west corner solution of the example of the three warehouses, of cost 440
- In each tableau the basic cells (blue) contain the amount $\bar{x}_{ij}$ in bold, and the other cells the reduced cost $\bar{c}_{ij}$
- The last column and the last row contain the potentials
- The entering cell is red, and the cells of the cycle are marked $\oplus$ and $\ominus$

Example: the potential method from the north-west corner (1/13)

In the first iteration the potentials of the north-west basis come from $\pi_7 = 0$, along the tree:

$$\pi_2 = \pi_7 + c_{27} = 2, \quad \pi_3 = \pi_7 + c_{37} = 6, \quad \pi_5 = \pi_2 - c_{25} = -2,$$

$$\pi_6 = \pi_2 - c_{26} = 0, \quad \pi_1 = \pi_5 + c_{15} = 7, \quad \pi_4 = \pi_1 - c_{14} = 6.$$

For example,

$$\bar{c}_{35} = c_{35} - \pi_3 + \pi_5 = 1 - 6 - 2 = -7.$$

Example: the potential method from the north-west corner (2/13)

Iteration 1

	4	5	6	7	π_i
1	**10**	**20**	-5	-4	7
2	7	**5** $\ominus$	**15**	**10** $\oplus$	2
3	9	-7 $\oplus$	-2	**30** $\ominus$	6
π_j	6	-2	0	0	

- The negative reduced costs are $\bar{c}_{35} = -7$, $\bar{c}_{16} = -5$, $\bar{c}_{17} = -4$, $\bar{c}_{36} = -2$; the cell $(3, 5)$ enters the basis.

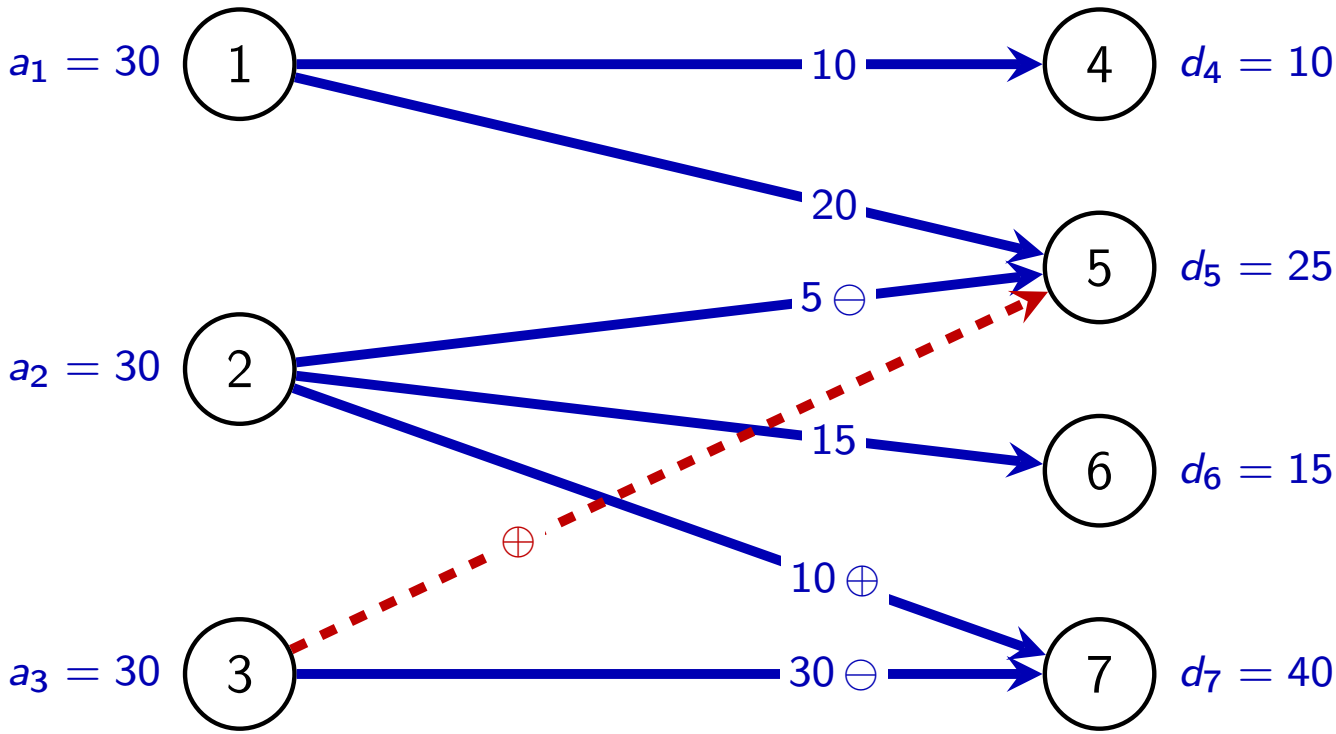
Example: the potential method from the north-west corner (3/13)

Iteration 1:

- The cycle is $(3,5)^+,\ (3,7)^-,\ (2,7)^+,\ (2,5)^-$.
- The cells with sign $-$ have $\bar{x}_{37} = 30$ and $\bar{x}_{25} = 5$: $\theta = 5$, and the cell $(2,5)$ leaves the basis.
- The cost goes from 440 to $440 + 5 \cdot (-7) = 405$.

Example: the potential method from the north-west corner (4/13)

In the network, the entering arc $(3, 5)$ (red, dashed) closes the cycle $3, 5, 2, 7, 3$ with the tree.



Example: the potential method from the north-west corner (5/13)

In the second iteration the cycle has six cells and visits all three rows.

Iteration 2

	4	5	6	7	π_i
1	**10**	**20** $\ominus$	-12 $\oplus$	-11	14
2	14	7	**15** $\ominus$	**15** $\oplus$	2
3	16	**5** $\oplus$	-2	**25** $\ominus$	6
π_j	13	5	0	0	

- The negative reduced costs are $\bar{c}_{16} = -12$, $\bar{c}_{17} = -11$, $\bar{c}_{36} = -2$; the cell $(1, 6)$ enters the basis.

Example: the potential method from the north-west corner (6/13)

Iteration 2:

- The cycle is $(1,6)^+,\ (1,5)^-,\ (3,5)^+,\ (3,7)^-,\ (2,7)^+,\ (2,6)^-$.
- The cells with sign $-$ have $\bar{x}_{15} = 20$, $\bar{x}_{37} = 25$ and $\bar{x}_{26} = 15$: $\theta = 15$, and the cell $(2,6)$ leaves the basis.
- The cost goes from 405 to $405 + 15 \cdot (-12) = 225$.

Example: the potential method from the north-west corner (7/13)

Iteration 3

	4	5	6	7	π_i
1	**10**	**5** $\ominus$	**15**	-11 $\oplus$	14
2	14	7	12	**30**	2
3	16	**20** $\oplus$	10	**10** $\ominus$	6
π_j	13	5	12	0	

- The only negative reduced cost is $\bar{c}_{17} = -11$: the cell $(1, 7)$ enters the basis.

Example: the potential method from the north-west corner (8/13)

Iteration 3:

- The cycle is $(1,7)^+,\ (1,5)^-,\ (3,5)^+,\ (3,7)^-$.
- The cells with sign $-$ have $\bar{x}_{15} = 5$ and $\bar{x}_{37} = 10$: $\theta = 5$, and the cell $(1,5)$ leaves the basis.
- The cost goes from 225 to $225 + 5 \cdot (-11) = 170$.

Example: the potential method from the north-west corner (9/13)

Iteration 4

	4	5	6	7	π_i
1	**10**	11	**15** $\ominus$	**5** $\oplus$	3
2	3	7	1	**30**	2
3	5	**25**	-1 $\oplus$	**5** $\ominus$	6
π_j	2	5	1	0	

- The only negative reduced cost is $\bar{c}_{36} = -1$: the cell $(3, 6)$ enters the basis.

Example: the potential method from the north-west corner (10/13)

Iteration 4:

- The cycle is $(3,6)^+,\ (3,7)^-,\ (1,7)^+,\ (1,6)^-$.
- The cells with sign $-$ have $\bar{x}_{37} = 5$ and $\bar{x}_{16} = 15$: $\theta = 5$, and the cell $(3,7)$ leaves the basis.
- The cost goes from 170 to $170 + 5 \cdot (-1) = 165$.

Example: the potential method from the north-west corner (11/13)

Iteration 5

	4	5	6	7	π_i
1	**10**	10	**10**	**10**	3
2	3	6	1	**30**	2
3	6	**25**	**5**	1	5
π_j	2	4	1	0	

- All the reduced costs are non-negative: the basic feasible solution is optimal, with cost $z = 165$.

Example: the potential method from the north-west corner (12/13)

- The optimal solution is

$$\tilde{x}_{14} = 10, \quad \tilde{x}_{16} = 10, \quad \tilde{x}_{17} = 10,$$

$$\tilde{x}_{27} = 30, \quad \tilde{x}_{35} = 25, \quad \tilde{x}_{36} = 5, \qquad z(\mathrm{TP}) = 165.$$

- Every cell outside the basis has $\bar{c}_{ij} > 0$, so by the Theorem «optimality criterion» the optimum is unique.

Example: the potential method from the north-west corner (13/13)

- The potentials $\tilde{\pi} = (3, 2, 5, 2, 4, 1, 0)$ are optimal for the $D(\mathrm{TP})$ model, with value

  $$30\,(3+2+5) - (10\cdot 2 + 25\cdot 4 + 15\cdot 1 + 40\cdot 0) = 300 - 135 = 165.$$

  Gurobi returns the same optimal solution, and its dual values coincide with $\tilde{\pi}$.

- From the minimum cost solution the method needs a single pivot: that solution has the basis of iteration 4, and the entering cell $(3, 6)$, with $\theta = 5$, gives the optimum.

Degeneracy

- When a basic cell has value 0, a pivot may have $\theta = 0$
- We now run the potential method on an instance where the north-west corner rule gives a degenerate solution
- We see a degenerate pivot, and why the basic cell with value 0 must be kept

Example: a degenerate pivot (1/6)

We consider the instance

c_{ij}	4	5	6	a_i
1	1	4	7	30
2	6	2	5	25
3	3	9	5	30
d_j	30	30	25	85

- In the north-west corner rule the first cell exhausts row 1 and column 4 at once.

Example: a degenerate pivot (2/6)

- Only the row is crossed out, and the cell $(2, 4)$ enters T with the amount 0: the solution, of cost 250, is degenerate.

step	cell	amount	crossed out
1	$(1, 4)$	30	row 1
2	$(2, 4)$	0	column 4
3	$(2, 5)$	25	row 2
4	$(3, 5)$	5	column 5
5	$(3, 6)$	25	row 3, column 6

Example: a degenerate pivot (3/6)

Iteration 1

	4	5	6	π_i
1	**30**	7	14	-7
2	**0** $\ominus$	**25** $\oplus$	7	-2
3	-10 $\oplus$	**5** $\ominus$	**25**	5
π_j	-8	-4	0	

- The only negative reduced cost is $\bar{c}_{34} = -10$: the cell $(3, 4)$ enters the basis.
- The cycle is $(3, 4)^+$, $(3, 5)^-$, $(2, 5)^+$, $(2, 4)^-$.

Example: a degenerate pivot (4/6)

Iteration 1:

- The cells with sign $-$ have $\bar{x}_{35} = 5$ and $\bar{x}_{24} = 0$: $\theta = 0$, and the cell $(2, 4)$ leaves the basis.
- The pivot is degenerate: the solution and its cost $z = 250$ do not change, only the basis does.

In the second iteration the basis has changed, so the potentials have changed too: now $\bar{c}_{15} < 0$.

Example: a degenerate pivot (5/6)

Iteration 2

	4	5	6	π_i
1	**30** $\ominus$	-3 $\oplus$	4	3
2	10	**25**	7	-2
3	**0** $\oplus$	**5** $\ominus$	**25**	5
π_j	2	-4	0	

- The only negative reduced cost is $\bar{c}_{15} = -3$: the cell $(1, 5)$ enters the basis.
- The cycle is $(1, 5)^+$, $(1, 4)^-$, $(3, 4)^+$, $(3, 5)^-$.

Example: a degenerate pivot (6/6)

Iteration 2:

- The cells with sign $-$ have $\bar{x}_{14} = 30$ and $\bar{x}_{35} = 5$: $\theta = 5$, and the cell $(3, 5)$ leaves the basis.
- The cost goes from 250 to $250 + 5 \cdot (-3) = 235$.

Iteration 3

	4	5	6	π_i
1	**25**	**5**	4	3
2	7	**25**	4	1
3	**5**	3	**25**	5
π_j	2	-1	0	

All the reduced costs are non-negative: the basic feasible solution is optimal, with cost $z = 235$. It is unique, and Gurobi confirms it.

What the degenerate example shows

- The basic cell with value 0 must be kept in T: without it, the other four cells would not span the six nodes, and the potentials could not be computed
- The degenerate pivot of iteration 1 does not lower the cost, but it moves to a basis whose potentials reveal an improving cell

Summary (1/2)

Potentials:

- $\bar{c}_{ij} = 0$ on the basic cells and $\pi_{m+n} = 0$
- they are the dual variables of the $D(\mathrm{TP})$ model

Optimality:

- the method stops when every reduced cost is non-negative
- then the potentials are an optimal dual solution, with the same value

Summary (2/2)

Pivot:

- otherwise θ units move around the cycle of an entering cell
- the cost decreases by $\theta \, |\bar{c}_e|$

Degeneracy:

- with $\theta = 0$ only the basis changes
- the basic cells with value 0 stay in the basis

The assignment problem

Introduction: The assignment problem

- n workers must be assigned to n jobs, one job each, at minimum total cost
- It is a transportation problem with unit supplies and demands, but the potential method makes many degenerate pivots on it
- We prove the theorem of König and Egerváry on zeros and lines, and we use it in the Hungarian algorithm, which works directly on the cost matrix

Definition 4: assignment problem

There are n workers, the origins $P = \{1, 2, \ldots, n\}$, and n jobs, the destinations $D = \{n+1, n+2, \ldots, 2n\}$; c_{ij} is the cost of assigning worker i to job j.

*An **assignment** gives every worker exactly one job and every job to exactly one worker.*

*The **assignment problem** asks for an assignment of minimum total cost.*

- We use one binary variable x_{ij} for every pair $i \in P$, $j \in D$, equal to 1 if worker i gets job j.

The AP model

Using these variables, an Integer Linear Programming (ILP) model for the assignment problem reads as follows:

$$\min \quad \sum_{i \in P} \sum_{j \in D} c_{ij} x_{ij}$$

$$\text{subject to} \quad \sum_{j \in D} x_{ij} = 1, \qquad \forall i \in P,$$

$$\sum_{i \in P} x_{ij} = 1, \qquad \forall j \in D,$$

$$x_{ij} \in \{0, 1\}, \quad \forall i \in P, \ \forall j \in D.$$

This model is referred to as the AP model.

The AP model: objective function and constraints

Description of the objective function and constraints of the AP model:

- the objective function minimizes the total cost of the assignment;
- the first constraints give every worker exactly one job;
- the second constraints give every job to exactly one worker;
- finally, the last constraints define the binary variables of the model.

- An $n \times n$ matrix with entries in $\{0, 1\}$ and exactly one 1 in every row and in every column is a **permutation matrix**
- The feasible solutions of the AP model are the permutation matrices: there are $n!$ assignments

Proposition 5: the LP relaxation of the assignment model

The LP relaxation of the AP *model, with* $x_{ij} \geq 0$ *in place of* $x_{ij} \in \{0, 1\}$*, is the* TP *model with* $a_i = 1$ *and* $d_j = 1$*. Its vertices are permutation matrices, and it has an optimal solution that is an assignment.*

Proof

- The constraints of the AP model are the supply and demand constraints of the TP model with unit supplies and demands, and the instance is balanced, with total n
- By the Proposition «feasibility and integrality» it has an optimal solution at a vertex, and every vertex is integer by the Theorem «integer vertices» of Chapter 4
- An integer feasible solution has $0 \leq x_{ij} \leq 1$, since x_{ij} is at most the sum of its row: it is a permutation matrix

□

Example: four workers and four jobs

Four workers $1, 2, 3, 4$ and four jobs $5, 6, 7, 8$ have the costs

c_{ij}	5	6	7	8
1	2	4	3	5
2	4	8	1	1
3	2	6	9	6
4	1	9	8	6

- The $4! = 24$ assignments, enumerated, have a unique minimum: worker 1 gets job 7, worker 2 job 8, worker 3 job 6, worker 4 job 5, with cost $3 + 1 + 6 + 1 = 11$.
- Gurobi gives the same assignment as the optimal solution of the LP relaxation.

Degeneracy of the potential method

- A basis of the LP relaxation has $2n - 1$ cells, but an assignment has only n cells with $x_{ij} = 1$
- Every basic feasible solution has $n - 1$ basic cells with value 0, and it is degenerate
- The potential method solves the LP relaxation, but many of its pivots have $\theta = 0$

Example: the north-west corner basis of the assignment instance

On the example of the four workers the north-west corner rule gives the basis below, with three basic zeros:

$\bar{x}_{ij}$	5	6	7	8	a_i
1	1				1
2	0	1			1
3		0	1		1
4			0	1	1
d_j	1	1	1	1	

From it the potential method reaches the optimum after 6 pivots, 4 of them degenerate.

Reductions and potentials

- The Hungarian algorithm works on the cost matrix: it subtracts constants from rows and columns
- These subtractions are potentials: the reduced matrix has entries $\bar{c}_{ij} = c_{ij} - \pi_i + \pi_j$
- The next lemma shows that they do not change the optimal assignments

Lemma 2: reduced costs of an assignment

For potentials $\boldsymbol{\pi}$, let $\bar{c}_{ij} = c_{ij} - \pi_i + \pi_j$. For every $\boldsymbol{x}$ that satisfies the constraints of the AP *model on rows and columns,*

$$\sum_{i \in P} \sum_{j \in D} c_{ij}\, x_{ij} = \sum_{i \in P} \sum_{j \in D} \bar{c}_{ij}\, x_{ij} + \sum_{i \in P} \pi_i - \sum_{j \in D} \pi_j .$$

Hence subtracting a constant from a row or from a column of the cost matrix changes the cost of every assignment by the same constant, and the optimal assignments do not change.

Proof

- It is the identity «cost and reduced cost» of the potential method with $a_i = d_j = 1$
- Subtracting r from row i gives the reduced costs of $\pi_i = r$ and the other potentials 0
- Subtracting s from column j gives those of $\pi_j = -s$

□

Corollary 1: zero-cost assignments are optimal

If $\bar{c}_{ij} \geq 0$ for every cell and an assignment uses only cells with $\bar{c}_{ij} = 0$, then it is optimal, even for the LP relaxation, its cost is $\sum_{i \in P} \pi_i - \sum_{j \in D} \pi_j$, and $\boldsymbol{\pi}$ is optimal for the dual of the LP relaxation, the $D(\mathrm{TP})$ model with $a_i = d_j = 1$.

Proof

- By the Lemma «reduced costs of an assignment», the assignment costs $\sum_{i \in P} \pi_i - \sum_{j \in D} \pi_j$
- Every $\boldsymbol{x} \geq 0$ satisfying the constraints on rows and columns costs at least as much, since $\bar{c} \geq 0$
- The potentials satisfy $\pi_i - \pi_j \leq c_{ij}$ with the same value, so they are optimal by weak duality



Definition 5: independent zeros and covers

Let $\bar{c}$ be an $n \times n$ matrix with non-negative entries.

*A set of zero entries is **independent** if no two of them lie in the same row or in the same column.*

*A set of **lines**, rows and columns, **covers** the zeros if every zero entry lies in at least one of the lines.*

Zeros, matchings and covers

- n independent zeros form an assignment of reduced cost zero, optimal by the Corollary «zero-cost assignments are optimal»

- In the bipartite network whose arcs are the cells with $\bar{c}_{ij} = 0$:
  - a set of independent zeros is a matching, a set of arcs without common endpoints as in Chapter 5 «Maximum flow»,
  - a cover is a set of nodes that touches every arc

Theorem 6: König and Egerváry

The maximum number of independent zeros of a matrix equals the minimum number of lines that cover all its zeros.

Proof (1/3)

Let M be a maximum set of independent zeros. A row or a column is **matched** if it contains a zero of M.

1. *Lines $\geq$ zeros.* The zeros of M lie in different rows and columns, so a line contains at most one of them, and a cover needs at least $|M|$ lines.

2. *Alternating paths.* An **alternating path** starts at an unmatched row, goes from a row to a column through a zero not in M, and from a column to a row through the zero of M of that column. Let R and K be the rows and the columns reached by alternating paths, the unmatched rows included.

Proof (2/3)

3. *Every column of K is matched.* Otherwise an alternating path ends at an unmatched column. Along it, the zeros not in M are one more than the zeros of M: exchanging them gives an independent set with $|M| + 1$ zeros, against the maximality of M.

4. *The lines.* Let L be the rows not in R and the columns in K. A zero (i, j) with row $i \in R$ has column $j \in K$:
   - if it is not in M, a path to i continues to j;
   - if it is in M, row i is matched, so it was reached from its column j.

   Hence L covers all the zeros.

Proof (3/3)

5. *Counting.* The rows not in R are matched, since the unmatched rows are in R, and the columns of K are matched.

   A zero of M in a column of K has its row in R, so it lies on exactly one line of L.

   Lines and zeros of M correspond one to one, and $|L| = |M|$.

□

- The proof is constructive: it gives the lines of a minimum cover from a maximum set of independent zeros, and a set that is not maximum is enlarged along an alternating path ending at an unmatched column

Algorithm: The Hungarian algorithm

Input : an $n \times n$ cost matrix c_{ij}, $i \in P$, $j \in D$
Output : an optimal assignment and optimal potentials $\boldsymbol{\pi}$

1 **Step 0 (reduction).** Subtract from each row its minimum, π_i; then subtract from each column of the result its minimum, $-\pi_j$. Let $\bar{c}$ be the reduced matrix.
2 **Step 1 (independent zeros).** Find a maximum set M of independent zeros of $\bar{c}$.
3 If $|M| = n$, `STOP`: M is an optimal assignment.
4 **Step 2 (lines).** Cover the zeros with $|M| < n$ lines, as in the proof of the theorem of König and Egerváry.
5 **Step 3 (adjustment).** Let δ be the minimum uncovered entry. Subtract δ from the uncovered entries and add it to the entries covered twice: $\pi_i := \pi_i + \delta$ for every uncovered row, $\pi_j := \pi_j + \delta$ for every covered column. Go to Step 1.

Rule of the examples and history

- **Rule of the examples.** In Step 1 the set M is built row by row, in increasing order: each row looks for an alternating path to an unmatched column, depth first, trying the columns in increasing order.

- The algorithm is due to H. W. Kuhn (1955), who named it after the Hungarian mathematicians D. König and J. Egerváry.

Theorem 7: correctness of the Hungarian algorithm

At every iteration $\bar{c}_{ij} = c_{ij} - \pi_i + \pi_j \geq 0$ *for every cell, and Step 3 increases the dual value* $\sum_{i \in P} \pi_i - \sum_{j \in D} \pi_j$ *by* $\delta\,(n - k) > 0$, *where* $k < n$ *is the number of lines.*

With integer costs the algorithm stops after a finite number of iterations with an optimal assignment, and $\boldsymbol{\pi}$ *is optimal for the dual of the LP relaxation.*

Proof (1/2)

1. *Step 0.* Subtracting the minimum of a row leaves non-negative entries and a zero in the row; the column minima are then ≥ 0, and subtracting them keeps the entries non-negative. By construction $\bar{c}_{ij} = c_{ij} - \pi_i + \pi_j$.

2. *Step 3.* All zeros are covered, so $\delta > 0$, and the uncovered entries stay ≥ 0. The update of the potentials gives:
   - $-\delta$ on uncovered row and column,
   - $-\delta + \delta = 0$ on uncovered row and covered column,
   - $+\delta$ on covered row and column,
   - 0 on covered row and uncovered column:

   exactly the adjustment.

Proof (2/2)

3. *Dual value.* With k_r covered rows and k_c covered columns, the value grows by

$$\delta\,(n - k_r) - \delta\,k_c = \delta\,(n - k),$$

and $k = |M| < n$ by the theorem of König and Egerváry.

4. *Termination.* By the Lemma «reduced costs of an assignment» the dual value never exceeds the cost of an assignment. With integer costs every δ is an integer ≥ 1, so the number of adjustments is finite. At the stop M is an assignment of zeros, optimal with π by the Corollary «zero-cost assignments are optimal».

□

Example: the Hungarian algorithm (1/9)

We run the algorithm on the example of the four workers.

Step 0. The row minima are $2, 1, 2, 1$, so $\pi_1 = 2$, $\pi_2 = 1$, $\pi_3 = 2$, $\pi_4 = 1$, and the rows become

$c_{ij} - \pi_i$	5	6	7	8
1	0	2	1	3
2	3	7	0	0
3	0	4	7	4
4	0	8	7	5

Example: the Hungarian algorithm (2/9)

- The column minima are $0, 2, 0, 0$: only column 6 changes, and $\pi_6 = -2$
- The dual value $2 + 1 + 2 + 1 + 2 = 8$ is a lower bound on the cost of every assignment

In each of the following matrices:

- the independent zeros are boxed,
- the lines are blue,
- the cell of δ is red,
- the potentials are on the margins.

Example: the Hungarian algorithm (3/9)

Iteration 1

$\bar{c}_{ij}$	5	6	7	8	π_i
1	0	$\boxed{0}$	1	3	2
2	3	5	$\boxed{0}$	0	1
3	$\boxed{0}$	2	7	4	2
4	0	6	7	5	1
π_j	0	-2	0	0	

- A maximum set of independent zeros (boxed) has 3 elements: $(1,6)$, $(2,7)$, $(3,5)$.
- The $3 < 4$ lines (blue) are the rows 1 and 2 and the column 5.

Example: the Hungarian algorithm (4/9)

Iteration 1:

- The minimum uncovered entry is $\delta = 2$, in the cell $(3, 6)$ (red).
- We subtract it from the uncovered entries and add it to the entries covered twice: π_i grows by 2 for the uncovered rows 3 and 4 and π_j grows by 2 for the covered column 5.
- The dual value grows by $2 \cdot (4 - 3) = 2$.

Example: the Hungarian algorithm (5/9)

Iteration 2

$\bar{c}_{ij}$	5	6	7	8	π_i
1	2	$\boxed{0}$	1	3	2
2	5	5	$\boxed{0}$	0	1
3	$\boxed{0}$	0	5	2	4
4	0	4	5	3	3
π_j	2	-2	0	0	

- A maximum set of independent zeros (boxed) has 3 elements: $(1,6)$, $(2,7)$, $(3,5)$.
- The $3 < 4$ lines (blue) are the row 2 and the columns 5 and 6.

Example: the Hungarian algorithm (6/9)

Iteration 2:

- The minimum uncovered entry is $\delta = 1$, in the cell $(1, 7)$ (red).
- We subtract it from the uncovered entries and add it to the entries covered twice: π_i grows by 1 for the uncovered rows 1, 3 and 4 and π_j grows by 1 for the covered columns 5 and 6.
- The dual value grows by $1 \cdot (4 - 3) = 1$.

Example: the Hungarian algorithm (7/9)

Iteration 3

$\bar{c}_{ij}$	5	6	7	8	π_i
1	2	0	$\boxed{0}$	2	3
2	6	6	0	$\boxed{0}$	1
3	0	$\boxed{0}$	4	1	5
4	$\boxed{0}$	4	4	2	4
π_j	3	-1	0	0	

- A maximum set of independent zeros (boxed) has 4 elements: $(1,7)$, $(2,8)$, $(3,6)$, $(4,5)$.
- They form an assignment of zero reduced cost: the algorithm stops, and the assignment is optimal.

Example: the Hungarian algorithm (8/9)

- The optimal assignment is

$$1 \to 7, \quad 2 \to 8, \quad 3 \to 6, \quad 4 \to 5,$$

with cost $3 + 1 + 6 + 1 = 11$, the unique optimum of the example of the four workers

- The final potentials $\pi_1 = 3$, $\pi_2 = 1$, $\pi_3 = 5$, $\pi_4 = 4$ and $\pi_5 = 3$, $\pi_6 = -1$, $\pi_7 = 0$, $\pi_8 = 0$ are an optimal dual solution, of value

$$(3 + 1 + 5 + 4) - (3 - 1 + 0 + 0) = 11.$$

- The dual value grew from 8 to 10 and then to 11

Example: the Hungarian algorithm (9/9)

The lines are those of the proof of the theorem of König and Egerváry.

- In iteration 1 row 4 is unmatched
- From it an alternating path reaches column 5 through its zero, and then row 3, matched to column 5
- Hence $R = \{3, 4\}$, $K = \{5\}$, and the lines are the rows 1, 2 and the column 5

Summary

The assignment problem:

- it is a transportation problem with unit supplies and demands
- the potential method makes many degenerate pivots on it

The Hungarian algorithm:

- it keeps $\bar{c}_{ij} = c_{ij} - \pi_i + \pi_j \geq 0$
- it covers the zeros with the minimum number of lines (König–Egerváry)
- it raises the dual value until n independent zeros give an optimal assignment

Typical mistakes

Introduction: Typical mistakes

- We collect the mistakes that are made most often with the transportation and the assignment problems
- They concern the balance of the instance, the bases, the potentials, the pivot and the Hungarian algorithm
- Check them before solving the exercises

Typical mistakes (1/4)

Observation 2: Typical mistakes

- ***Solving an unbalanced instance as it is.*** *With more supply than demand the equality constraints have no solution: add a dummy destination with zero costs, or a dummy origin when the demand exceeds the supply.*
- ***Dropping a basic zero.*** *A degenerate basis still has $m + n - 1$ cells; without the cell of value 0 the tree is disconnected and the potentials cannot be computed.*

Typical mistakes (2/4)

Observation: Typical mistakes

- ***Taking a set with a cycle as a basis.*** *$m + n - 1$ cells are a basis only if they contain no closed path alternating rows and columns.*
- ***Mixing the sign conventions.*** *Here $\bar{c}_{ij} = c_{ij} - \pi_i + \pi_j$ with $\pi_i - \pi_j = c_{ij}$ on the basic cells; with $u_i + v_j = c_{ij}$ the destination potentials change sign.*

Typical mistakes (3/4)

Observation: Typical mistakes

- ***Computing θ on all the cells of the cycle.*** *θ is the minimum on the cells with sign $-$ only; the entering cell receives θ.*
- ***Covering the zeros with too many lines.*** *The lines of the Hungarian algorithm must be a minimum cover, as many as the independent zeros; with more lines δ may be wrong or the stop missed.*

Typical mistakes (4/4)

Observation: Typical mistakes

- ***Forgetting the entries covered twice.*** *In Step 3 of the Hungarian algorithm δ is subtracted from the uncovered entries and added to those covered by a row and a column.*
- ***Running the algorithms on profits.*** *They minimize: with profits p_{ij} to maximize, use the costs $c_{ij} = -p_{ij}$, or $c_{ij} = p_{\max} - p_{ij}$ with $p_{\max}$ the largest profit, to keep them non-negative.*

Summary

Before solving:

- balance the instance with a dummy origin or destination
- turn profits into costs

Potential method:

- keep the basic zeros, and never take a set with a cycle as a basis
- one sign convention; θ only on the cells with sign $-$

Hungarian algorithm:

- a minimum cover, and δ added to the entries covered twice

Summary (1/3)

- The TP model ships the supplies a_i of m origins to the demands d_j of n destinations; it is an MCF model on the complete bipartite network, feasible exactly when it is balanced
- An uncapacitated transshipment instance becomes a transportation instance with the distances as costs
- A basis is a spanning tree of the bipartite network, with $m + n - 1$ cells; its basic solution is found leaf by leaf and is integer for integer data

Summary (2/3)

- The north-west corner and the minimum cost rules build a basic feasible solution, one line per step
- The potential method: potentials with $\bar{c}_{ij} = 0$ on the basis and $\pi_{m+n} = 0$; optimal when $\bar{c} \geq 0$; otherwise θ units move around the cycle of the entering cell, and the cost drops by $\theta\,|\bar{c}_e|$
- The AP model is a TP model with unit data; the Hungarian algorithm reduces rows and columns, covers the zeros with a minimum number of lines (König–Egerváry) and adjusts by δ, raising the dual value

Summary (3/3)

The instances at a glance.

warehouses and stores	north-west 440, minimum cost 170, optimum 165
potential method	4 pivots from north-west, 1 from minimum cost
optimal potentials	$\tilde{\boldsymbol{\pi}} = (3, 2, 5, 2, 4, 1, 0)$, with $\pi_7 = 0$
with $a_1 = 40$	dummy destination 8, optimum 155
degenerate instance	$\theta = 0$, then $\theta = 5$; optimum 235
transshipment, no capacities	optimum 49
workers and jobs	bound 8, $\delta = 2$, $\delta = 1$, optimum 11